OPTIMAL INQUIRY

Tai-Wei Hu and Andriy Zapechelnyuk

Abstract. A decision maker acquires and processes information about an uncertain state of nature through an inquiry: a contingent sequence of questions ending in a decision. Inquiries are costly, with cost proportional to their length. We characterize optimal inquiries, which determine both which decisions are considered and which are prioritized, in the sense of being reached after fewer questions. We show that optimal inquiries generate endogenous confirmation: relative to the no-cost benchmark, they expand the set of states in which the decision maker chooses the most likely decisions. An optimal inquiry may also treat ex ante symmetric options differently, prioritizing some while deprioritizing or excluding others. To an observer who ignores inquiry costs, these choices may appear biased.

JEL Classification: D81, D83.

Keywords: Bounded rationality, information theory, confirmation bias, consideration set

Date: August 24, 2026

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We are grateful to Olivier Compte, Francesc Dilmé, Matthew Elliott, Federico Echenique, Philippe Jehiel, Benson Leung, Michael Mandler, John Moore, Larry Samuelson, Jakub Steiner, Colin Stewart, Ina Taneva, Tymon Tatur, Chris Tyson, Peter Wagner, and Zaifu Yang, as well as anonymous referees and many conference and seminar participants, for helpful comments.

1. Introduction

Many important decisions are made through inquiry. A doctor examines a patient by asking various questions, ordering tests, and deciding when enough has been learned to treat, refer, or hospitalize. A criminal investigation proceeds by questioning witnesses and suspects, and deciding when the evidence is sufficient to act. In such settings, recent work suggests that inaccurate decisions often arise not merely from a lack of information, but also from the procedure by which information is gathered, sequenced, and evaluated.[1] Moreover, since inquiries are costly, both in terms of cognitive effort and physical resources, the procedure itself becomes part of the decision problem. We ask what an optimal inquiry looks like when these costs are taken into account, and how the resulting procedure can generate patterns that may appear, from an ex post perspective, as mistakes or biases.

This paper studies a model of costly inquiry. An inquiry is a strategy for asking questions about a relevant state of nature. It starts with an initial question and a contingent plan that specifies which question to ask depending on the answers to previous questions. The cost of inquiry is directly associated with asking questions and processing their answers. More questions allow the decision maker to make more accurate decisions, but they also increase the cost of the procedure. An optimal inquiry balances these two considerations.

Our analysis focuses on the structure of the procedure: which questions are asked, in what order, and which actions are taken after different histories of answers. By attaching the cost directly to the steps of the decision procedure, our model differs from models that represent information acquisition through an ex ante cost of experiments. In our setting, two inquiries that induce the same final information partition may nevertheless have different values because they reach decisions through different paths. We characterize optimal inquiry by separating its dynamic outcome into two components: its form and its content. The form consists of the set of actions reached with positive probability, which we call the *consideration set*, and the number of questions required to reach each action. The content is the information partition induced by the questions, where each cell is called a *category* and leads to an action in the consideration set.

[1]See, for example, Singh et al. (2017) and Croskerry et al. (2013) for misdiagnosis in healthcare and Gould and Leo (2010) for wrongful convictions in criminal investigations.

Our main result gives two interconnected principles that determine the optimal inquiry. First, we show that the optimal form determines the content: once the consideration set and the number of questions required to reach each action are fixed, the optimal information partition emerges endogenously, by assigning each action a subset of states in which this action has the highest payoff net of inquiry costs. Thus, conditional on the form, there is no further problem of information design. Second, we show a negative correlation between the ex ante probability of choosing an action and the number of questions needed to reach that action. That is, more likely actions are prioritized and reached after fewer questions.

The above two principles lead to our main behavioral implication, *endogenous confirmation*. By this we mean that the optimal procedure expands the categories associated with actions that are reached after fewer questions, thereby increasing their choice probability. Since optimal inquiry assigns shorter paths to actions that are more likely to be chosen, the procedure reinforces—and in this sense *confirms*—the actions that it prioritizes. From the perspective of an outside observer who ignores inquiry costs, this may look like a mistake or a bias toward the most likely actions.[2] The effect is *endogenous* because both the prioritized actions and the extent of their expanded choice regions are determined by the primitives of the decision problem and the cost of inquiry. We show that every optimal inquiry exhibits endogenous confirmation.

This feature is especially striking when all feasible actions are ex ante symmetric. Without inquiry costs, choice probabilities are evenly distributed across actions. With costly inquiry, however, the decision maker may optimally break this symmetry by reaching some actions earlier, expanding their choice regions, or dropping other actions from the consideration set altogether. Such preferential treatment may appear biased to an outside observer, but it arises solely from the incentive to economize on inquiry costs. Our results illustrate that the procedure of information processing is itself an object of economic choice. Changes in the economic environment, such as the cost of inquiry, can therefore change which actions are prioritized and which apparent biases emerge.

[2]In this sense, this feature is related to what the psychology literature calls confirmation bias, broadly understood as "unwitting selectivity in the acquisition and use of evidence" (Nickerson, 1998). We emphasize, however, that our notion is a property of the optimal inquiry, not an assumed psychological bias. It may therefore appear as a "bias" only from the perspective of an outside observer who ignores inquiry costs.

Related Literature. This paper makes a conceptual and methodological contribution to three strands of literature.

The first strand studies procedural approaches to information processing and decision making. Wilson (2014) formulates the decision-making process as a finite automaton. The main result in Wilson (2014) is a dynamic-consistency type of result called multi-self consistency, with the cognitive constraint modelled through an exogenously given memory capacity. In contrast, our model is dynamically consistent in the conventional sense, and the size of the optimal inquiry is endogenous. Glazer and Rubinstein (2014) study questionnaires as economic procedures, but in a principal–agent model. Our focus is different: there is no strategic respondent, and the decision maker designs the inquiry to balance decision accuracy against the cost of asking and processing questions. Cremer et al. (2007) propose a model of organizational language using codes, with the main trade-off between the use of broader codes, which are easier to process, and the precision of such codes. While our model shares a similar trade-off, we focus on the procedural nature of information processing, with implications for timing.

Similar to our paper, Mandler (2024) proposes a model in which the decision maker acquires information by asking questions modelled as a partition of states. The research question addressed in Mandler (2024) is complementary to ours. The focus of Mandler (2024) is on the implementation of an exogenously given decision rule at minimum cost, with implications on how the inquiries should be structured. In contrast, our model features an endogenous decision rule that is jointly determined with the inquiry tree and information structure, which allows us to show the connection between cognitive costs and behavioral biases.

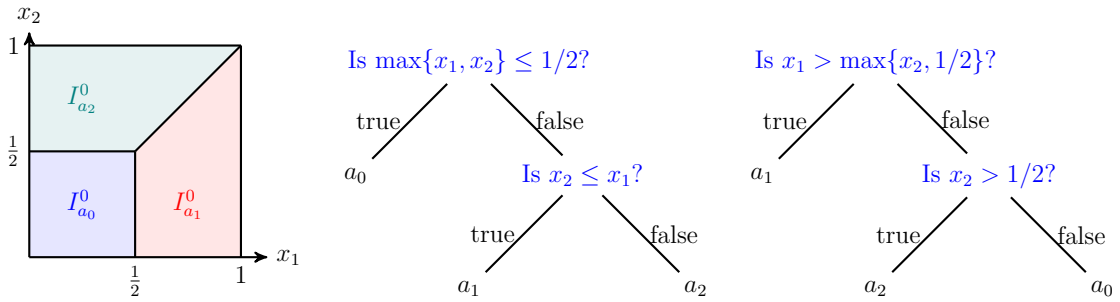
The second strand of literature includes papers that study behavioral biases with cognitive frictions. These papers range from axiomatic to constrained optimization approaches, the former including Masatlioglu et al. (2012) and Manzini and Mariotti (2014) and the latter including Caplin et al. (2019). While our paper is closer to the latter, our characterization also delivers properties familiar from the axiomatic literature, such as dynamic consistency and the attention-filter property of Masatlioglu et al. (2012). Our endogenous confirmation is related to, but differs from the literature that rationalizes confirmation bias. The wisdom from the literature is that frictions in information processing tend to cause a decision maker

to favor signals that confirm the prior belief. Wilson's (2014) model generates this form of confirmation bias based on limited memory, but in her model the decision maker only passively processes it. In contrast, in our model confirmation arises through the endogenous design of the inquiry procedure: the decision maker chooses the questions, their order, and the resulting information partition. Steiner et al. (2017) obtain a "status quo bias" in a dynamic rational inattention model where the decision maker tends to stick to actions that are optimal ex ante. Nimark and Sundaresan (2019) and Novák et al. (2024) show that players adopt signal structures that are determined by and reinforce their prior beliefs. All these papers argue that certain implications from the proposed models can be interpreted as confirmation bias and emphasize the importance of the prior belief. Jehiel and Steiner (2020) obtain confirmation bias in a model where the decision maker chooses whether or not to continue to receive more signals, but can only remember the last one received. Confirmation bias here means that the agent is more likely to stop when seeing a signal in favor of the prior. In contrast, endogenous confirmation in our model arises from an endogenous inquiry structure and emphasizes the procedural nature of information processing.

The third strand is the rational inattention literature initiated by Sims (2003), where agents can choose flexibly what information to acquire, often modeled as a choice from a broad class of Blackwell experiments; see Maćkowiak et al. (2023). In the same spirit, our approach allows for arbitrary content of a question. The most closely related paper in this strand is Saint-Paul (2017), which studies deterministic choice under a discrete relative-entropy cost, with information processed through a discrete partition similar to our categories. Unlike this literature, however, we model information processing as a concrete inquiry procedure: the cost is attached directly to asking questions and processing their answers, rather than to an ex ante information structure. As a result, the ordering of questions matters and has behavioral implications.

2. Example

We illustrate the main ideas with a simple example of misdiagnosis in primary health care. A patient visits a doctor. The patient's health condition is summarized by a state $x = (x_1, x_2) \in [0,1]^2$, where x_1 represents the diagnostic difficulty of the case and x_2 represents its seriousness. The doctor has three possible actions: a medication prescription (labeled



a. Optimal partition b. Inquiry I c. Inquiry II

FIGURE 1. Optimal information partition and optimal inquiries for $\lambda = 0$

a_0), a referral to a specialist (labeled a_1), and hospitalization (labeled a_2). The doctor's utility from these actions in state x is given by

$$u(a_0, x) = 0, \ u(a_1, x) = 2x_1 - 1, \ u(a_2, x) = 2x_2 - 1. \tag{1}$$

Thus, a_1 is best in difficult cases, a_2 is best in serious cases, and a_0 is best otherwise.

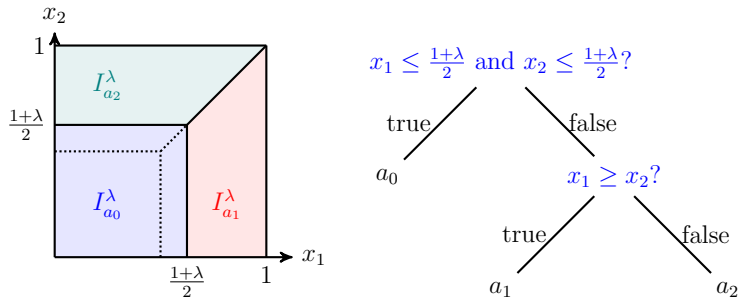
The doctor is initially uninformed about x. To learn about x, the doctor follows an *inquiry*. An inquiry is a strategy of how to ask questions. It starts with an initial question, specifies follow-up questions depending on earlier answers, and eventually prescribes an action. Each question asks whether a proposition about x is true or false; these can be propositions about different dimensions of x or propositions about the relationship across different dimensions.

A cost λ is deducted from the doctor's utility whenever she asks a question. This cost represents the opportunity cost of time and cognitive effort that could instead be spent diagnosing and treating other patients. If the doctor reaches an action a after asking ℓ questions, the resulting payoff is $u(a, x) - \lambda\ell$. The doctor chooses an inquiry to maximize her expected utility net of the cost, given her prior belief about x. Assume that this prior belief has full support on $[0, 1]^2$ and admits a density.

As a benchmark, suppose that there is no cost of asking questions, $\lambda = 0$. Observe that, by (1), for each $a \in \{a_0, a_1, a_2\}$, it is optimal to choose a if and only if $x \in I^0_a$, where

$$I^0_{a_0} = \{x : \max\{x_1, x_2\} \leq 1/2\}, \quad I^0_{a_1} = \{x : \max\{x_1, x_2\} > 1/2 \text{ and } x_1 > x_2\},$$

$$I^0_{a_2} = \{x : \max\{x_1, x_2\} > 1/2 \text{ and } x_1 \leq x_2\},$$



a. Optimal partition b. Optimal inquiry

FIGURE 2. Optimal information partition and optimal inquiry for $\lambda > 0$

as illustrated in Figure 1a. We refer to $I^0 = \{I^0_{a_0}, I^0_{a_1}, I^0_{a_2}\}$ as an *information partition*. When $\lambda = 0$, all inquiries that induce I^0 are payoff-equivalent. To anticipate the role of positive costs, consider inquiries that induce I^0 with the fewest number of questions. In general, there are several such inquiries. For illustration, consider inquiries I and II shown in Figures 1b and 1c. These inquiries are equally good in achieving the goal of inducing the optimal partition I^0.

Suppose that, under the zero-cost benchmark, prescribing medication is the ex ante most likely correct choice, that is, $\mathbb{P}[x \in I^0_{a_0}] > \max\{\mathbb{P}[x \in I^0_{a_1}], \mathbb{P}[x \in I^0_{a_2}]\}$. Now suppose that questions are costly, $\lambda > 0$. Then, inquiries I and II are no longer equally good. Although they induce the same partition I^0, inquiry I does so at strictly lower expected cost. This is because inquiry I asks only one question in the most likely event, $x \in I^0_{a_0}$, while inquiry II asks two questions in the same event. The motive to save inquiry cost leads to the *prioritization* of the most likely action a_0, in the sense that the optimal inquiry poses fewer questions to reach a_0 and more questions to reach a_1 or a_2.

In addition, the doctor may find it optimal to sacrifice some accuracy in the decision to reduce the cost of inquiry; in other words, the optimal information partition is endogenously determined by the cost. Indeed, for a range of λ's that are not too high, the optimal inquiry prioritizes action a_0. Taking the inquiry cost into account, optimal choices are now determined by comparing the utilities of the actions net of the costs: $u(a_0, x) - \lambda$, $u(a_1, x) - 2\lambda$, and $u(a_2, x) - 2\lambda$. By (1), this comparison results in the optimal information partition $I^\lambda = \{I^\lambda_{a_0}, I^\lambda_{a_1}, I^\lambda_{a_2}\}$ shown in Figure 2a. The associated optimal inquiry is shown in Figure 2b.

As seen from the comparison of Figures 1a and 2a, as λ increases, the doctor optimally expands the set of states $I^{\lambda}_{a_0}$ that lead to the higher-priority action a_0, and shrinks the sets where lower-priority actions a_1 and a_2 are taken. We refer to this effect as *endogenous confirmation*.

By *confirmation*, we mean that the doctor optimally expands the region of states in which action a_0 is taken, making it easier for the procedure to confirm the ex ante most plausible hypothesis. An outside observer who does not take the inquiry cost into account may classify the doctor's decisions at states in $I^{\lambda}_{a_0} - I^{0}_{a_0}$ (Figure 2a, the area between the dotted and solid lines) as "mistakes": in these states, the doctor prescribes medication although the zero-cost benchmark would prescribe referral to a specialist or hospitalization.

This effect is *endogenous* because which action the doctor prioritizes in the optimal inquiry and confirms to is endogenously determined by the cost of inquiry and the underlying distribution of the states. For example, suppose that $\mathbb{P}[x \in I^{0}_{a_1}] > \mathbb{P}[x \in I^{0}_{a_0}]$. Then, it is possible that the optimal inquiry prioritizes a_1 for small λ but prioritizes a_0 for intermediate values of λ, because the cost-saving benefit of expanding $I^{\lambda}_{a_0}$ may dominate that from prioritizing a_1.[3]

The rest of the paper characterizes this logic for arbitrary finite action sets and general state spaces.

3. Model

3.1. **Primitives.** A decision maker (DM) needs to process information about an uncertain state of nature before taking an action. The DM's utility $u(a, x)$ depends on her action, $a \in A$, and the state, $x \in X$. The set of actions A is finite and contains at least two actions. The set of states X is a convex subset of $\mathbb{R}^L$, $L \in \mathbb{N}$. State x is distributed according to a probability distribution G that is absolutely continuous and has full support on X. We will use notation $\mathbb{P}[\cdot]$ and $\mathbb{E}[\cdot]$ to denote the probability and expectation under G, respectively. Throughout the paper, we assume:

- (A_1) For all $a \in A$, $u(a, x)$ is continuous in x, and $\mathbb{E}[u(a, x)]$ is finite.
- (A_2) For all distinct $a', a'' \in A$ and all $c \in \mathbb{R}$, the set $\{x \in X : u(a', x) - u(a'', x) = c\}$ has measure zero.

[3]This is the case, for example, when x_1 and x_2 are i.i.d. under truncated exponential distribution such that the probabilities $\mathbb{P}[x \in I^{0}_{a_1}]$ and $\mathbb{P}[x \in I^{0}_{a_0}]$ are similar enough.

Assumption (A_1) is needed for the DM's optimization problem to be well defined. Assumption (A_2) means that the utility curves of any two actions are almost never parallel to each other. Most utility functions that naturally emerge in applications satisfy (A_1) and (A_2).[4]

3.2. **Inquiries.** When confronted with a state x, the DM does not observe x directly. Instead, to obtain information about x, she relies on an inquiry: a series of true/false questions formulated as propositions. A proposition is a statement about x in the form "$x \in Y$" that can be either true or false. We denote the collection of Borel subsets of X by $\mathcal{B}(X)$, and identify a proposition with a set $Y \in \mathcal{B}(X)$. We say that proposition Y is *true* if $x \in Y$ and it is *false* if $x \notin Y$.

An *inquiry* $Q = \langle N, T, \sigma, \chi, d \rangle$ is a finite binary tree. Non-terminal nodes of the tree are associated with propositions, and terminal nodes are associated with actions. Specifically:

- a finite set N of nodes contains a root n^o and a nonempty set T of terminal nodes (note that the tree may consist of a single terminal node, i.e., $N = T = \{n^o\}$);
- each non-terminal node $n \in N - T$ is followed by two edges labelled *true* and *false*;
- successor function σ assigns to each non-terminal node $n \in N - T$ and each edge $e \in \{true, false\}$ a child $\sigma(n, e) \in N$ of node n following edge e;
- proposition mapping χ assigns to each non-terminal node $n \in N - T$ a proposition $\chi(n) \in \mathcal{B}(X)$;
- decision rule d assigns to each terminal node $t \in T$ an action $d_t \in A$.

We denote by $\mathcal{Q}_X$ the set of all possible inquiries given a set of states X.

Given a state $x \in X$, an inquiry $Q = \langle N, T, \sigma, \chi, d \rangle$ begins with the proposition $\chi(n^o)$ at the root. At each non-terminal node $n \in N - T$, the inquiry asks whether it is true that $x \in \chi(n)$. If true, then the inquiry proceeds to the node $\sigma(n, true)$; otherwise, it proceeds to the node $\sigma(n, false)$. When a terminal node $t \in T$ is reached, the DM takes action d_t.

Each inquiry Q *categorizes* states into subsets through a series of questions. Given the answers to the questions leading to a terminal node $t \in T$, the DM infers that the true state belongs to a subset of states $I_t(Q)$. We refer to $I_t(Q)$ as the *category* associated with

[4]In fact, we impose (A_2) for notational convenience. It guarantees that, conditional on the form, the optimal content is essentially unique as indifferences can only occur with probability zero. This assumption can be substantially relaxed at the expense of more cumbersome notation.

terminal node t. The collection of categories $\{I_t(Q) : t \in T\}$ forms a partition of X. It is the information partition at the end of the inquiry.

As zero probability events do not matter for payoffs, we adopt and use throughout the paper a measure-based notion of partition that disregards sets of measure zero under G. Specifically:

Definition 3.1. A collection of disjoint sets $\{X_1, ..., X_K\}$ is a *partition of* X if $\sum_{k=1}^{K} \mathbb{P}(X_k) = 1$.

Consistent with this definition, equalities and inclusions involving categories or unions of categories are understood up to G-null sets.

3.3. **Payoffs.** We assume that asking questions is costly. Let the DM's cost of any single question be $\lambda > 0$. Given an inquiry Q, let $\ell_t(Q)$ be the length of the path from n^o to t in the tree, that is, $\ell_t(Q)$ is the number of questions asked to reach terminal node t. Then, the cost of inquiry at terminal node t is equal to $\lambda \ell_t(Q)$.

We can now formulate the DM's optimization problem. Given an inquiry Q and a state x, if the inquiry reaches the terminal node t, the DM's ex post payoff net of the cost is $u(d_t, x) - \lambda \ell_t(Q)$. Because each terminal node $t \in T$ is reached whenever the state x is in $I_t(Q)$, the DM's ex ante expected utility is given by

$$W(Q; \lambda) = \sum_{t \in T} \int_{x \in I_t(Q)} \Big(u(d_t, x) - \lambda \ell_t(Q) \Big) G(\mathrm{d}x). \tag{2}$$

An inquiry Q is *optimal* if it solves

$$\max_{Q \in \mathcal{Q}_X} W(Q; \lambda). \tag{3}$$

3.4. **Outcomes.** We now show that an optimal inquiry can be summarized by its payoff-relevant outcome, which separates the form of the inquiry tree from the informational content induced at its terminal nodes. The key observation is that an optimal inquiry cannot assign the same action to two distinct terminal nodes. As a result, each terminal node corresponds to a unique action in A.

Lemma 3.1. *If an inquiry $Q = \langle N, T, \sigma, \chi, d \rangle$ is optimal, then $d_t \neq d_{t'}$ for all pairs of distinct terminal nodes $t, t' \in T$.*

Indeed, if two terminal nodes led to the same action, we could construct another inquiry that preserves the same action chosen in every state while reducing the number of questions on some paths. This is easy to see if the two terminal nodes were adjacent as they could be directly merged; the construction is slightly more complicated in other cases but follows a similar principle. Based on Lemma 3.1, from now on, we identify terminal nodes with actions they induce.

Given an inquiry Q, let D be the set of actions associated with the terminal nodes of Q. We refer to D as the *consideration set*.[5] Intuitively, only actions in the consideration set are considered by the inquiry, and one can reduce inquiry costs by shrinking this set. For each action $d \in D$, let ℓ_d be the length of inquiry leading to the terminal node where d is chosen, and let I_d be the category induced by Q in that terminal node. Let $\ell = (\ell_d)_{d\in D}$ be the *length profile* and let $I = (I_d)_{d\in D}$ be the *information partition* induced by Q. We will refer to the triple (D, ℓ, I) as the *outcome* induced by Q. The pair (D, ℓ) captures the shape of the inquiry tree, and hence is referred to as the *form* of Q. The partition I captures the informational content in the terminal nodes of the inquiry tree, and hence is referred to as the *content* of Q.

An outcome (D, ℓ, I) captures all we need to know to evaluate the DM's expected payoff of an inquiry that leads to that outcome. Indeed, for any two different inquiries Q and Q' that induce the same outcome (D, ℓ, I), the expected utility in (2) implies that they deliver the same ex ante payoff to the DM.

An outcome (D, ℓ, I) is *implementable* if there exists an inquiry that induces it. The following lemma characterizes implementable outcomes.

Lemma 3.2. *An outcome (D, ℓ, I) is implementable if and only if*

$$\sum_{d\in D} 2^{-\ell_d} = 1. \tag{4}$$

Lemma 3.2 implies that implementability of an outcome depends only on the form (D, ℓ), through the Kraft equality (4); the content I does not affect implementability as long as it

[5]We adopt this terminology from Masatlioglu et al. (2012). Our optimal inquiry satisfies a minimal rationality property for the consideration set required by Masatlioglu et al. (2012): independence of irrelevant alternatives, or the "attention filter". It is defined as follows. Suppose that D is a strict subset of A. The attention filter property holds if, for any smaller action set $A' \subset A$ that contains D, the optimal consideration set is still D.

has one category for each action in D. Equality (4) follows from the Kraft inequality (Kraft, 1949) in information theory that characterizes the path lengths of binary trees.[6] For any form (D, ℓ) satisfying (4) and any partition I with $|D|$ categories, the proof constructs an inquiry that induces the corresponding outcome. This lemma also fully characterizes possible length configurations for a given consideration set. For example, with $|D| = 3$, only three length profiles satisfy equality (4), namely, $\ell = (1, 2, 2)$, $\ell = (2, 1, 2)$, and $\ell = (2, 2, 1)$. With $|D| = 4$, there are 13 length profiles that satisfy (4), namely, the uniform profile, $(2, 2, 2, 2)$, and 12 distinct permutations of the profile $(1, 2, 3, 3)$.

3.5. **Modelling assumptions.** Our assumption of binary questions can be motivated by the following interpretation of the decision-making procedure. In any optimal inquiry, each terminal node is associated with a distinct decision, and each node in the inquiry tree partitions the terminal nodes into two subsets. Since terminal nodes correspond to actions, the associated question can be interpreted as helping the DM determine which subset of actions contains the optimal action. In such a procedure, binary questions are a natural benchmark, and our model endogenizes both the partition of the remaining actions at each stage and the criterion used to choose between the two groups. This interpretation is natural in settings such as diagnosis, investigation, procurement, or other institutional decision processes in which the DM must progressively narrow down the relevant alternatives. It is less suitable for parts of information acquisition that occur before any decision-relevant alternatives have been formulated.

The cost of inquiry should also be interpreted broadly. Depending on the application, it may represent cognitive effort, time pressure, or physical resources used to obtain and process evidence. What is essential for the model is that each step of the inquiry consumes scarce resources before some alternatives can be ruled out. We assume that each question has the same cost and that total inquiry cost is linear in the number of questions. This is mainly a benchmark assumption. It isolates the procedural trade-off between decision accuracy and inquiry length and allows us to characterize optimal inquiry analytically. The assumption also has two implicit restrictions: the cost of a question does not depend on its content, and the cost of a question does not depend on when it is asked. The first

[6]Here we have equality instead of inequality because in our inquiry trees each non-terminal node has precisely two outgoing branches.

restriction can be interpreted as a fixed cost of making a binary distinction at each stage; the second as a constant opportunity cost of continuing the inquiry. More general models with heterogeneous or time-dependent question costs would change the coding problem, but the present benchmark keeps the central mechanism transparent.

4. Optimal Inquiries

We have shown that an inquiry can be summarized by its outcome (D, ℓ, I). Moreover, Lemma 3.2 shows that the information partition I does not affect whether or not an outcome is implementable. This characterization allows us to solve the optimal inquiry problem in two stages. We first fix an arbitrary form (D, ℓ) that satisfies (4), and solve for the optimal content $I = I^*(D, \ell)$. Then, we maximize over all possible forms (D, ℓ).

In the first stage, taking (D, ℓ) as given, we find an information partition $I^*(D, \ell)$ that maximizes the DM's expected utility. Specifically, let $I^*(D, \ell) = \{I_d^*(D, \ell)\}_{d \in D}$, where

$$I_d^*(D, \ell) = \Big\{x \in X : u(d, x) - \lambda \ell_d > \max_{a \in D - \{d\}} u(a, x) - \lambda \ell_a\Big\}. \tag{5}$$

That is, for each action $d \in D$, $I_d^*(D, \ell)$ is the set of states where d is the unique best-response action among all actions in D when the DM takes into account the cost of inquiry associated with each action.[7] Thus, conditional on (D, ℓ), the DM weakly prefers $I^*(D, \ell)$ to any content I. Formally, using the notation $W(D, \ell, I; \lambda)$ for the DM's expected utility from an outcome (D, ℓ, I), we have

$$\begin{aligned} W(D, \ell, I^*(D, \ell); \lambda) &= \int_{x \in X} \left(\max_{d \in D}(u(d, x) - \lambda \ell_d)\right) G(\mathrm{d}x) \\ &\geq \sum_{d \in D} \int_{x \in I_d} \Big(u(d, x) - \lambda \ell_d\Big) G(\mathrm{d}x) = W(D, \ell, I; \lambda). \end{aligned} \tag{6}$$

This observation is a key to solving the optimal inquiry problem. The outcomes of an inquiry include both a continuous element I and a discrete element (D, ℓ). The optimal content I is determined by the form (D, ℓ) through $I^*(D, \ell)$, and problem (3) can now be reduced to the choice of the form, (D, ℓ).

Let $\mathcal{F}^*$ be the set of all forms (D, ℓ) with nonempty $D \subseteq A$ and ℓ satisfying (4). We have the following theorem.

[7] Note that $I^*(D, \ell)$ is a partition of X according to Definition 3.1, because, by assumption (A_2), the set $\big(X - \bigcup_{d \in D} I_d^*(D, \ell)\big)$ has measure zero. Moreover, if $D = \{d\}$, then define $I_d^*(D, \ell) = X$.

Theorem 4.1. *An inquiry Q is optimal if and only if its form (D, ℓ) solves*

$$\max_{(D,\ell)\in\mathcal{F}^*} \int_{x\in X} \left(\max_{d\in D} (u(d,x) - \lambda \ell_d) \right) G(\mathrm{d}x), \tag{7}$$

and its content I is identical to $I^*(D,\ell)$ up to a measure zero set.

Theorem 4.1 implies three properties of optimal inquiry.

First, an optimal inquiry exists, since $\mathcal{F}^*$ and D are finite sets, and the integral in (7) is well defined by assumption (A_1).

The second property concerns the optimal consideration set. Let A^* contain all the actions which are uniquely optimal under some $x \in X$:

$$A^* = \left\{ a \in A : \ u(a,x) > \max_{a'\in A-\{a\}} u(a',x) \text{ for some } x \in X \right\}.$$

That is, each $a \in A^*$ is uniquely optimal on an open set of states, and each $a' \notin A^*$ is strictly inferior to some $a \in A^*$ in each state except, possibly, a measure zero of states. For λ sufficiently small, the optimal consideration set must remain $D = A^*$, because any D that omits actions from A^* creates a strictly positive loss in decision payoff, while any potential saving in inquiry cost is proportional to λ and hence negligible for small λ.[8] For λ sufficiently large, the optimal consideration set D contains a single action, because the cost of asking any question outweighs its potential benefit. This is summarized in the next corollary.

Corollary 4.1. *There exist two thresholds $\lambda_2 > \lambda_1 > 0$ such that for all $\lambda < \lambda_1$, the optimal consideration set is $D = A^*$; and for all $\lambda > \lambda_2$, every optimal consideration set is given by $D = \{a\}$ with $a \in \arg\max_{a\in A} \mathbb{E}[u(a,x)]$.*

The third property concerns prioritization. We say that action d is *prioritized* relative to action d' if $\ell_d < \ell_{d'}$, that is, d is reached after fewer questions than d'. Under optimal inquiry, more likely actions are always prioritized over less likely actions (that is, more likely actions are reached after weakly fewer questions), as the next corollary shows.

Corollary 4.2. *Let Q be an optimal inquiry and let (D, ℓ, I) be its outcome. For all $d, d' \in D$, if $\mathbb{P}(I_d) > \mathbb{P}(I_{d'})$, then $\ell_d \leq \ell_{d'}$.*

[8]The other case, where $D \supsetneq A^*$, is slightly different: in that case we can remove the extraneous actions so that the expected length at $\lambda = 0$ is strictly lower and for small but positive λ, this leads to a strictly higher overall payoff.

Indeed, if under some inquiry an action d is more likely than an action d' and yet it takes more questions to reach d, then the DM would be better off by maintaining the same information partition but swapping the roles of d and d' in the inquiry tree; this swap is possible due to Lemma 3.2. This property is, in fact, related to a well-known result in computer science called *Huffman coding*, an algorithm that minimizes the average inquiry lengths for a given probability distribution $(\mathbb{P}(I_d))_{d\in D}$ subject to the constraint (4) (see, e.g., Cover and Thomas, 2006, Section 5). The key distinctive feature of our model is that the probabilities used as input for the Huffman coding are endogenously determined. Building on Corollary 4.2, in the next section, we show how the cost of inquiry affects the ex ante choice probability of prioritized actions, the endogenous confirmation effect.

5. Endogenous Confirmation

We now study how the cost of inquiry affects the content of an optimal outcome. We use two approaches. The first approach compares the outcomes of the optimal inquiry against those of the zero-cost benchmark, and the second approach considers the comparative statics as the cost increases.

5.1. Comparison with the zero-cost benchmark.

Using the first approach, we show that the DM's choices under optimal inquiry exhibit *endogenous confirmation*. Informally, this means that the DM chooses one of her most likely actions with a greater probability than what she would optimally do if information was costless. To formalize this notion, we rank the actions according to their likelihood under the optimal inquiry. We thus define the "most likely actions" from the DM's perspective, and this captures the DM's subjective assessment of the likelihood of which actions tend to be optimal. Let Q be an inquiry, let (D, ℓ, I) be the outcome induced by Q, and let $K = |D|$. Let us order elements in $D = \{d_k\}_{k=1}^K$ so that

$$\begin{gathered}\mathbb{P}(I_{d_1}) \geq \mathbb{P}(I_{d_2}) \geq ... \geq \mathbb{P}(I_{d_K}),\\ \text{with a tie-breaking rule } \ \mathbb{P}(I_{d_k}) = \mathbb{P}(I_{d_{k+1}}) \implies \ell_{d_k} \leq \ell_{d_{k+1}}.\end{gathered} \tag{8}$$

We now define endogenous confirmation. For each $k = 1, ..., K$, let $E_k^Q = \bigcup_{t=1}^k I_{d_t}$ be the set of states that lead to an action in $\{d_1, ..., d_k\}$ (i.e., one of k most likely actions) under the inquiry Q. Also, let $E_k^0 = \bigcup_{t=1}^k I_{d_t}^0$ be the set of states that lead to an action

in $\{d_1, ..., d_k\}$ under the zero-cost benchmark.[9] The difference $E_k^Q - E_k^0$ is the set of states where a "mistake" is made: inquiry Q leads to one of the k most likely actions when the zero-cost benchmark prescribes otherwise. In any of these states, the DM can be seen as "biased" towards her most likely actions by an outside observer who does not take the cost of information into account.

Definition 5.1. An inquiry Q with outcome (D, ℓ, I) has *endogenous confirmation* if for all orders $(d_k)_{k=1}^K$ that satisfy (8) and all $k \in \{1, 2, ..., K\}$,

$$E_k^Q \supseteq E_k^0. \tag{9}$$

It has *strict endogenous confirmation* if, in addition, there exists an order that satisfies (8) and $k \in \{1, 2, ..., K\}$ such that $E_k^Q - E_k^0$ has a positive measure.[10]

This expansion of the categories associated with the most likely actions can have two margins. On the one hand, when $D \neq A^*$, so D is not the optimal consideration set under the zero-cost benchmark, there is an extensive margin in that the cost induces the DM to optimally drop some actions from consideration. In this case, if there is endogenous confirmation, then it is always strict, because under positive cost the choice is in D with probability one, but under zero cost the choice is outside of D with positive probability. On the other hand, if $D = A^*$, then only the intensive margin is at work, and strict endogenous confirmation would require a non-trivial expansion to occur for k most likely actions for some $k < K$.

Given a form (D, ℓ), we say that the length profile ℓ is *uniform* if the length of inquiry leading to each action in D is the same, namely, $\ell_d = \log_2 |D|$ for all $d \in D$. Note that this can only happen if $|D| = 2^k$ for some $k \in \mathbb{N}$.

Theorem 5.1. *For every $\lambda > 0$, every optimal inquiry has endogenous confirmation. Moreover, it has strict endogenous confirmation, unless its consideration set D is optimal under $\lambda = 0$ and its length profile ℓ is uniform.*

[9]For each $d \in A$, I_d^0 is the set of all states where action d is the best, as defined by (5) with $\lambda = 0$.

[10]The definition of endogenous confirmation is easily generalizable to include mixed strategies and non-partitional information structures (Hu and Zapechelnyuk, 2024, Section 4.3). Using this definition, we can compare our results to other literature. Specifically, one can ask if endogenous confirmation holds under rational inattention or not, and the general answer is no. See Hu and Zapechelnyuk (2024, Section 6) for more details.

Endogenous confirmation of the optimal inquiry emerges for the following reason. From Corollary 4.2 we know that the most likely actions are associated with shorter inquiry lengths. At the same time, given the optimal length profile, the optimal information partition given by (5) has the feature that actions associated with shorter inquiry lengths will be chosen on a larger set of states relative to the zero-cost benchmark. These two factors reinforce each other and give rise to endogenous confirmation.

Theorem 5.1 uses the ordering of actions in Definition 5.1, which is based on their likelihood under the optimal inquiry rather than under the zero-cost benchmark. This ordering is appropriate for our purpose because endogenous confirmation concerns the implications of the inquiry cost faced by the decision maker. Moreover, the optimality of actions used under the zero-cost benchmark may turn out to be fragile: they may no longer be worthwhile to consider if the benefit of differentiating them from other actions is low compared to the cost.

A related implication of Theorem 5.1 is that the probability distribution over actions induced by the optimal inquiry first-order stochastically dominates that induced by the zero-cost benchmark, since (9) implies

$$\sum\nolimits_{d\in\{d_1,...,d_k\}} \mathbb{P}(I_d) \geq \sum\nolimits_{d\in\{d_1,...,d_k\}} \mathbb{P}(I_d^0), \quad \text{for each } k = 1, ..., K, \tag{10}$$

and the inequality is strict for some k if endogenous confirmation is strict.

5.2. **Comparative statics.** Now we turn to the second approach and consider the comparative statics as λ increases. Specifically, for a given cost $\lambda > 0$, consider a neighborhood of costs, $[\lambda - \epsilon, \lambda + \epsilon]$, such that some form (D, ℓ) is optimal for all costs in this interval. For each cost in that neighborhood, let $Q^{\lambda'} = (D, \ell, I^{\lambda'})$ be the optimal inquiry under λ'. Our focus is how the optimal content $I^{\lambda'}$ changes as the cost λ' increases within the neighborhood where the optimal form (D, ℓ) is constant. Note that the definition takes into account the possibility that there may be multiple optimal forms.

Definition 5.2. We say that endogenous confirmation is *locally increasing at* λ if there is an $\epsilon > 0$ such that (a) there is a constant form that is optimal over $[\lambda-\epsilon, \lambda+\epsilon]$, and (b) for every optimal form (D, ℓ) that remains optimal over $[\lambda-\epsilon, \lambda+\epsilon]$ and for any fixed associated order $(d_k)_{k=1}^K$ satisfying (8) under the associated optimal inquiry Q^λ, for all $k \in \{1, ..., K-1\}$, $E_k^{Q^{\lambda'}}$ is increasing for all $\lambda' \in [\lambda - \epsilon, \lambda + \epsilon]$ w.r.t. the set inclusion order, and endogenous

confirmation is *locally strictly increasing at* λ if, in addition, $E_k^{Q^{\lambda'}}$ is strictly increasing for some k.

The following proposition shows that the identified endogenous confirmation locally intensifies with λ, in the sense that, whenever the optimal form (D, ℓ) remains constant, the DM optimally makes more "errors" and becomes more biased toward the actions that are most likely under (D, ℓ).

Proposition 5.1. *For generic λ—that is, for an open dense set of costs—endogenous confirmation is locally increasing. Moreover, it is locally strictly increasing, unless the length profile of every optimal form at λ is uniform.*

The next argument is immediate by Proposition 5.1. Let N be the number of actions that are optimal under the zero-cost benchmark, $N = |A^*|$. Suppose that we can strictly rank actions in $A^* = \{a_1, ..., a_N\}$ according to the choice probability under zero cost, i.e., $\mathbb{P}(I^0_{a_1}) > ... > \mathbb{P}(I^0_{a_N})$. Except for nongeneric ties in the Huffman-coding problem, it uniquely determines the form of the optimal inquiry. For λ small, the ranking under optimal inquiry is preserved. Proposition 5.1 then implies that, for each $k = 1, ..., N$, $\sum_{m=1}^{k} \mathbb{P}(I^\lambda_{a_m})$ locally increases with λ (and strictly so for some k if the associated length profile is not uniform). In other words, the most likely actions under the zero-cost benchmark will continue to be the most likely for small λ, and in fact, their likelihood increases with the cost.

To conclude this section, we recall our example in Section 2. As mentioned there, under the assumption that $\mathbb{P}(I^0_{a_0}) > \max\{\mathbb{P}(I^0_{a_1}), \mathbb{P}(I^0_{a_2})\}$, the optimal inquiry prioritizes action a_0 for a range of λ's that are not too large, and the doctor first inquires about the optimality of prescribing medication before investigating the optimality of referrals or emergency treatment. Theorem 5.1 and Proposition 5.1 then imply that the doctor optimally expands the category of states in which a_0 is chosen, as shown in Figure 2a. Moreover, since we show that endogenous confirmation holds in general in our model, the details of the example, such as the underlying distribution of states and the detailed specification of the utility functions, do not matter for the main conclusion.

6. Further Implications

We have characterized optimal inquiries and shown how inquiry costs generate endogenous confirmation. We now provide several further properties of optimal inquiry. These properties illustrate how costly inquiry affects not only the final information partition, but also the structure of the decision procedure itself.

6.1. **Dynamic consistency and commitment.** Optimal inquiries are dynamically consistent in that the DM could not profitably deviate from the inquiry at any stage of the process if she was given the opportunity. As a result, it makes no difference whether the DM commits to an optimal inquiry ex ante or she is free to update her strategy at any interim stage. The proof uses the typical argument for dynamic consistency. Crucial to this argument is the additive realized-path cost structure, which makes the continuation problem separable from the preceding history. With nonseparable cost structures, such as costs based on the maximum depth of the inquiry tree, dynamic consistency can fail.

6.2. **Endogenous prioritization of some actions in ex ante symmetric situations.** Here we show that the cost of inquiry may lead the DM to prioritize some choices over others, even when all options are ex ante symmetric. One example is a choice from several ex ante symmetric products. There is a set $\{a_1, ..., a_N\}$ of products whose values $(x_1, ..., x_N)$ are uncertain and follow an i.i.d. distribution. The DM chooses one of the products and receives its value. Thus, $A = \{a_1, ..., a_N\}$, $X = \mathbb{R}^N_+$, and $u(a_n, x_1, ..., x_N) = x_n$ for each n. This implies that $\mathbb{P}(I^0_a) = \mathbb{P}(I^0_{a'})$ for all $a, a' \in A$. We have the following result.

Corollary 6.1. *Suppose that all actions are ex ante equally likely under zero cost,* $\mathbb{P}(I^0_a) = 1/|A|$ *for all* $a \in A$.
(i) If $|A| \neq 2^k$ *for any* $k \in \mathbb{N}$, *then for all* $\lambda > 0$ *there exist* $d, d' \in A$ *such that* $\mathbb{P}(I^\lambda_d) > \mathbb{P}(I^\lambda_{d'})$.
(ii) If $|A| = 2^k$ *for some* $k \in \mathbb{N}$, *then there exists* $\lambda_1 > 0$ *such that the following holds. For all* $\lambda > \lambda_1$ *there exist* $d, d' \in A$ *such that* $\mathbb{P}(I^\lambda_d) > \mathbb{P}(I^\lambda_{d'})$.

We prove Corollary 6.1 by showing that, as the inquiry cost increases, the DM eventually reduces the average inquiry length. This can be achieved in two ways. First, the DM may have a reduced consideration set to shorten the inquiry tree. Second, within the consideration set, more likely actions will be endogenously prioritized so that they are selected after

fewer questions in the inquiry process, and thus become even more likely due to endogenous confirmation (Theorem 5.1), despite being ex ante symmetric.[11] Either of the two will imply that some actions are prioritized in terms of ex ante choice probabilities. We emphasize that this is not due to any conflict of interest but arises solely from the motive to save inquiry cost.

6.3. **Elimination of close actions from consideration.** Finally, actions in the optimal consideration set must be sufficiently different from one another. If two actions have uniformly similar payoffs, then the benefit of distinguishing between them is smaller than the cost of asking an additional question. The optimal inquiry therefore retains at most one of them.

Corollary 6.2. *If actions a' and a'' are such that $\sup_{x\in X}|u(a',x)-u(a'',x)|<\lambda$, then at most one of them belongs to the optimal consideration set.*

Taken together, these implications highlight the central message of the paper. Costly inquiry affects not only how much information is processed, but also the procedure by which information is organized and decisions are reached. Because the DM must economize on questions, the optimal inquiry may shorten the decision process, prioritize some actions over others, exclude payoff-similar alternatives, and expand the set of states assigned to actions reached earlier in the inquiry. Thus, phenomena that may appear as behavioral biases from the perspective of the zero-cost benchmark can arise endogenously from the optimal design of a costly decision procedure.

[11] If the DM randomizes over inquiries or actions in terminal nodes, there could exist an optimal symmetric mixed inquiry under the assumptions of Corollary 6.1. However, such a randomization requires indifference and a generic small perturbation of the primitives would break this indifference. In this sense, optimal randomization is a non-generic feature. Note that, although the symmetry assumption in Corollary 6.1 is itself non-generic, the asymmetry identified there is locally robust: for any fixed symmetric environment covered by Corollary 6.1, the resulting asymmetry is bounded away from zero and therefore continues to hold under sufficiently small perturbations.

Appendix A. Proofs

A.1. **Proof of Lemma 3.1.** Before proving Lemma 3.1, we provide three claims.

First, we show that, given an inquiry Q with $K+1$ terminal nodes, if we want to get rid of one specific terminal node, there is a way to construct an inquiry Q' such that K remaining terminal nodes are reached weakly faster and some strictly faster than under Q.

Claim A.1. *Let $K \geq 1$. If $\ell = (\ell_1, ..., \ell_{K+1}) \in \mathbb{N}^{K+1}$ satisfies $\sum_{k=1}^{K+1} 2^{-\ell_k} = 1$, then there exists $\ell' = (\ell'_1, ..., \ell'_K) \in \mathbb{N}^K$ such that $\ell'_k \leq \ell_k$ for all $k = 1, ..., K$, $\ell'_{k_0} < \ell_{k_0}$ for some $k_0 \in \{1, ..., K\}$, and $\sum_{k=1}^{K} 2^{-\ell'_k} = 1$.*

Proof. Let $\ell^o = \max_{k \leq K+1} \ell_k \geq 1$. Then there are two k_1, k_2 such that $\ell_{k_1} = \ell^o = \ell_{k_2}$: since $2^{\ell^o} = 2^{\ell^o} \sum_{k=1}^{K+1} 2^{-\ell_k} = \sum_{k=1}^{K+1} 2^{\ell^o - \ell_k}$ is even, the number of terms with $\ell_k = \ell^o$ must be even. We consider two cases.

(a) $K+1 = k_1$. Let $\ell'_k = \ell_k$ for all $k \neq k_1, k_2$ and let $\ell'_{k_2} = \ell_{k_2} - 1$. Thus,

$$\sum_{k=1}^{K} 2^{-\ell'_k} = \sum_{k \neq k_1, k_2} 2^{-\ell_k} + 2^{-\ell'_{k_2}} = \sum_{k \neq k_1, k_2} 2^{-\ell_k} + 2^{-\ell_{k_2}+1} = \sum_{k=1}^{K+1} 2^{-\ell_k} = 1,$$

where the last equality follows from $\ell_{K+1} = \ell_{k_2}$.

(b) Suppose that $\ell_{K+1} < \ell^o$. Then, consider the new vector $\ell'_k = \ell_k$ for all $k \neq k_1, k_2$, and $\ell'_{k_1} = \ell_{k_1} - 1 = \ell'_{k_2}$. The sum of the new vector is $\sum_{k=1}^{K} 2^{-\ell'_k} = \sum_{k \neq k_1, k_2, k \leq K} 2^{-\ell_k} + 2 * 2^{-\ell'_{k_2}} = \sum_{k=1}^{K} 2^{-\ell_k} + 2^{-\ell^o+1} = 1 - 2^{-\ell_{K+1}} + 2^{-\ell^o+1} \leq 1$, with equality if and only if $\ell^o = \ell_{K+1} + 1$. In that case, we are done. Now, if the inequality is strict, merge the two coordinates, k_1 and k_2, into a single one with length $\ell'_{k_1,k_2} = \ell_{k_1} - 1$. Note that for the new vector, $(\ell_{-\{k_1,k_2\}}, \ell'_{k_1,k_2}, \ell_{K+1})$, the Kraft equality is preserved. We can then repeat the same process. Note that at the end of the process we can keep track of the identities of the merged coordinates. We continue until ℓ_{K+1} is the longest and apply (a). To recover the full vector, if two vectors are merged into one, we add one more length to each so that the Kraft equality is preserved. If a merged coordinate originally came from two coordinates of length ℓ, and ends with length $\ell' \leq \ell - 1$, splitting it gives $\ell' + 1 \leq \ell$ for each original coordinate, and strictness is preserved. □

Second, we show that, given an information partition with K elements and a length profile that satisfies condition (11) below, there is a way to construct an inquiry with K terminal nodes such that its length profile and information partition are as given.

Claim A.2. *Let $K \geq 2$, let $I = (I_k)_{k=1}^K$ be a partition of X into K elements, and let $\ell = (\ell_k)_{k=1}^K \in \mathbb{N}^K$ be such that*

$$\sum_{k=1}^K 2^{-\ell_k} = 1. \tag{11}$$

Then, there exists an inquiry $Q = \langle N, T, \sigma, \chi, d\rangle$ with a set $T = \{t_1, ..., t_K\}$ of terminal nodes such that for all $k = 1, ..., K$,

$$\ell_{t_k}(Q) = \ell_k \quad \textit{and} \quad I_{t_k}(Q) = I_k. \tag{12}$$

Proof. By Theorem 5.2.1 in Cover and Thomas (2006), (11) implies that there exists a finite binary tree with a set of nodes N and a successor relation over N, with K terminal nodes labeled $t_1, ..., t_K$, such that, for each $k = 1, ..., K$, the length of the path from the root to each terminal node t_k is exactly ℓ_k.

We now construct an inquiry $Q = \langle N, T, \sigma, \chi, d\rangle$ that satisfies (12). Let N be as above, and let $T = \{t_1, ..., t_K\}$. Decision rule d is arbitrary. For each nonterminal node $n \in N - T$, let us associate two edges leading out of n with $true$ and $false$, and define the map σ so that $\sigma(n, true) = n^{true}$ if $n \rightsquigarrow n^{true}$ along the edge labelled $true$ and $\sigma(n, false) = n^{false}$ if $n \rightsquigarrow n^{false}$ along the edge labelled $false$.

It remains to construct a proposition mapping χ that yields the partition I in the terminal nodes. First, we associate each node $n \in N$ with a subset of states, $X_n \subseteq X$, as follows. For each $t_k \in T$, let $X_{t_k} = I_k$. Then, by backward induction, for each nonterminal node $n \in N - T$, let $X_n = X_{\sigma(n,true)} \cup X_{\sigma(n,false)}$. This implies that X_{n^o} at the root n^o is equal to X up to a measure-zero set, since $(I_k)_{k=1}^K$ is a partition, and we can place a measure-zero set anywhere in the propositions used along the tree without affecting the payoffs. Finally, define a proposition map χ as follows. For each nonterminal node $n \in N - T$, let $\chi(n) = X_{\sigma(n,true)}$. By induction from the root of the tree, it is straightforward to verify that this proposition mapping χ, together with the successor function σ, induces the required information partition I. □

Next, we show that every node of an optimal inquiry is reached with positive probability.

Claim A.3. *If an inquiry $Q = \langle N, T, \sigma, \chi, d\rangle$ is optimal, then every node $n \in N$ is reached with positive probability.*

Proof. Let $Q = \langle N, T, \sigma, \chi, d\rangle$ be an optimal inquiry. Suppose, by contradiction, that some $n' \in N$ is reached with probability zero, but all the predecessors are reached with positive probability. Let n be the immediate predecessor of n', and let n'' be the second successor of n. Consider now a new inquiry $\hat{Q}$ obtained by modifying Q as follows. First, eliminate all the nodes succeeding n' (including n'). Second, eliminate the branch that connects n and n'' and identify n with n'', so that all the nodes following n'' remain but for each terminal node t following n'', $\hat{\ell}(t)$ in $\hat{Q}$ is one less than $\ell(t)$ in Q. Note that for terminal nodes t in $\hat{Q}$, (11) holds. Moreover, since n' happens with zero-probability, for the set of terminal nodes inherited by $\hat{Q}$, denoted by $\hat{T}$, the collection $\{I_t(Q)\}_{t\in\hat{T}}$ forms a partition according to Definition 3.1. By Claim A.2 and its proof, we can construct the questions in $\hat{Q}$ so that the final partition is the same as $\{I_t(Q)\}_{t\in\hat{T}}$, up to measure-zero sets. Clearly, every terminal node $t \in T$ that is reached with positive probability under Q is reached with the same probability under $\hat{Q}$, and the DM's expected payoff conditional on reaching any such node is unchanged. But the length of inquiry for the terminal nodes in the branch that starts from n'' is uniformly shorter under $\hat{Q}$ and this happens with a positive probability. This contradicts the optimality of Q. □

We now prove Lemma 3.1. Let $Q = \langle N, T, \sigma, \chi, d\rangle$ be an optimal inquiry. Suppose, by contradiction, that $d_{t'} = d_{t''}$ for some $t', t'' \in T$ with $t' \neq t''$. Let $K = |T| - 1$, and let us label the terminal nodes consecutively, $T = \{t_1, ..., t_K, t_{K+1}\}$, such that $t_K = t'$ and $t_{K+1} = t''$. We assume $K \geq 2$ as the case $K = 1$ is obvious. Without loss we assume that $\ell_{t_K}(Q) \leq \ell_{t_{K+1}}(Q)$.

Now we construct an alternative inquiry, $Q' = (N', T', \sigma', \chi', d')$, with $|T'| = K$ terminal nodes that leads to a strictly higher expected value to the DM. Let

$$I'_k = I_{t_k}(Q) \text{ for each } k = 1, ..., K-1, \quad \text{and} \quad I'_K = I_{t_K}(Q) \cup I_{t_{K+1}}(Q), \tag{13}$$

and let $d'_k = d_{t_k}$ for each $k = 1, ..., K$.

Now, by the Kraft inequality (e.g., Cover and Thomas, 2006, Theorem 5.2.1), we have $\sum_{k=1}^{K+1} 2^{-\ell_{t_k}(Q)} = 1$. It is an equality instead of inequality, because in our inquiry tree every non-terminal node has exactly two branches. By Claim A.1, there exists $\ell' \in \mathbb{N}^K$ such that

$$\ell'_k \leq \ell_{t_k}(Q) \text{ for all } k = 1, ..., K, \ \ell'_k < \ell_{t_k}(Q) \text{ for some } k \in \{1, ..., K\}, \tag{14}$$

and $\sum_{k=1}^{K} 2^{-\ell'_k} = 1$. By Claim A.2 applied to $I' = \{I'_k\}_{k=1}^K$, $\ell' = (\ell'_1, ..., \ell'_K)$, and $d' = (d'_1, ..., d'_K)$, there exists an inquiry $Q' = \langle N', T', \sigma', \chi', d' \rangle$ with $T' = \{t_1, ..., t_K\}$ such that

$$I'_{t_k}(Q') = I'_k \quad \text{and} \quad \ell'_{t_k}(Q') = \ell'_k \quad \text{for all } k = 1, ..., K. \tag{15}$$

Thus, we obtain

$$\begin{aligned} W(Q'; \lambda) &= \sum_{k=1}^{K} \int_{I_{t_k}(Q')} (u(d'_k, x) - \lambda \ell'_{t_k}(Q')) G(\mathrm{d}x) = \sum_{k=1}^{K} \int_{I'_k} (u(d'_k, x) - \lambda \ell'_k) G(\mathrm{d}x) \\ &> \sum_{k=1}^{K+1} \int_{I_{t_k}(Q)} (u(d_{t_k}, x) - \lambda \ell_{t_k}(Q)) G(\mathrm{d}x) = W(Q; \lambda), \end{aligned}$$

where the second equality is by (15), and the inequality is by (13), (14), and that, by Claim A.3, all the terminal nodes in T are reached with positive probability under Q. □

A.2. **Proof of Lemma 3.2.** The necessity follows from Theorem 5.2.1 in Cover and Thomas (2006). We have an equality instead of inequality, because in our inquiry tree every non-terminal node has exactly two branches. The sufficiency follows from the existence of an inquiry $Q = \langle N, T, \sigma, \chi, d \rangle$ whose outcome is (D, ℓ, I) by Claim A.2. □

A.3. **Proof of Theorem 4.1.** The result that an inquiry is optimal if and only if its form solves (7) is immediate by equations (2) and (5), inequality (6), and the definition of $\mathcal{F}^*$ together with Lemma 3.2.

It remains to show that if (D, ℓ, I) is an optimal outcome, then I is identical to $I^*(D, \ell)$ up to a measure zero set. Let (D, ℓ) be given, and let $I^* = I^*(D, \ell)$. For any partition $I = \{I_d : d \in D\}$, let

$$W(D, \ell, I; \lambda) = \sum_{d \in D} \int_{I_d} [u(d, x) - \lambda \ell_d] G(\mathrm{d}x).$$

By (5), for any I and any $d \in D$, if $x \in I^*_d \cap I_{d'}$ with $d \neq d'$ then

$$u(d, x) - \lambda \ell_d > u(d', x) - \lambda \ell_{d'}.$$

Thus, since $\mathbb{P}(X - \cup_{d\in D} I_d^*) = 0$ by (A_2) and the fact that G has full support,

$$W(D,\ell,I^*;\lambda) - W(D,\ell,I;\lambda) = \sum_{d,d'\in D} \int_{I_d^*\cap I_{d'}} \left[(u(d,x)-\lambda\ell_d) - (u(d',x)-\lambda\ell_{d'})\right] G(\mathrm{d}x)$$

$$- \sum_{d\in D} \int_{I_d\cap(X-\cup_{d\in D} I_d^*)} [u(d,x)-\lambda\ell_d] G(\mathrm{d}x)$$

$$= \sum_{d\neq d'\in D} \int_{I_d^*\cap I_{d'}} \left[(u(d,x)-\lambda\ell_d) - (u(d',x)-\lambda\ell_{d'})\right] G(\mathrm{d}x) \geq 0,$$

and the inequality is strict if $\mathbb{P}(I_d^* \cap I_{d'}) > 0$ for some $d \neq d'$. □

A.4. **Proofs of Theorem 5.1 and Proposition 5.1.** We first provide an auxiliary lemma.

Fix $\lambda \in \mathbb{R}_+$ and $(D,\ell) \in \mathcal{F}^*$. Let $K = |D|$ and let the actions in D be ordered according to their lengths, so $D = \{d_k\}_{k=1}^K$ such that

$$\ell_{d_1} \leq \ell_{d_2} \leq ... \leq \ell_{d_K}. \tag{16}$$

Let X_k^λ be the event that an action in $\{d_1,...,d_k\}$ is strictly preferred to all actions in $\{d_{k+1},...,d_K\}$ when cost λ is taken into account:

$$X_k^\lambda = \left\{ x \in X : \max_{k'=1,...,k} u(d_{k'},x) - \lambda\ell_{d_{k'}} > \max_{m=k+1,...,K} u(d_m,x) - \lambda\ell_{d_m} \right\}. \tag{17}$$

Lemma A.1. *Let $\lambda > 0$ and $(D,\ell) \in \mathcal{F}^*$. Let $K = |D|$, and let the actions in $D = \{d_1,...,d_K\}$ be ordered according to (16). Suppose that both X_k^λ and $X - X_k^\lambda$ have positive measure for all $k = 1,...,K-1$. Then, for each $\lambda' \in [0,\lambda)$,*

$$X_k^{\lambda'} \subseteq X_k^\lambda \quad \text{for all } k = 1,2,...,K-1. \tag{18}$$

Moreover, if ℓ is not uniform, then there exists $k^* \in \{1,...,K-1\}$ such that

$$\ell_{d_{k^*}} < \ell_{d_{k^*+1}} \text{ and the set } X_{k^*}^\lambda - X_{k^*}^{\lambda'} \text{ has a positive measure.} \tag{19}$$

Proof. First, we prove (18). Let $k \in \{1,...,K-1\}$. Suppose by contradiction that there exists $x \in X_k^{\lambda'}$ such that $x \notin X_k^\lambda$. By (17), $x \in X_k^{\lambda'}$ and $x \notin X_k^\lambda$ imply that there exist k^*

and m^*, with $k^* \leq k < m^*$, such that

$$u(d_{k^*}, x) - \lambda' \ell_{d_{k^*}} = \max_{k'=1,...,k} u(d_{k'}, x) - \lambda' \ell_{d_{k'}} > u(d_{m^*}, x) - \lambda' \ell_{d_{m^*}}, \tag{20}$$

$$u(d_{k^*}, x) - \lambda \ell_{d_{k^*}} \leq \max_{k'=k+1,...,K} u(d_{k'}, x) - \lambda \ell_{d_{k'}} = u(d_{m^*}, x) - \lambda \ell_{d_{m^*}}. \tag{21}$$

Combining (20) and (21), we obtain

$$\lambda(\ell_{d_{m^*}} - \ell_{d_{k^*}}) \leq u(d_{m^*}, x) - u(d_{k^*}, x) < \lambda'(\ell_{d_{m^*}} - \ell_{d_{k^*}}).$$

This is a contradiction to $\lambda > \lambda'$ and $\ell_{d_{m^*}} \geq \ell_{d_{k^*}}$, where the last inequality is by (16) and $m^* > k^*$. This proves (18).

Next, suppose ℓ is not uniform. Then, for every order that satisfies (16), by definition of the uniform profile, there exists $k^* \in \{1, ..., K-1\}$ such that

$$\ell_{d_1} = ... = \ell_{d_{k^*}} < \ell_{d_{k^*+1}} \leq ... \leq \ell_{d_K}. \tag{22}$$

We now prove that the set $X^\lambda_{k^*} - X^{\lambda'}_{k^*}$ has a positive measure. Define

$$\bar{u}(x) = \max_{k=1,...,k^*} u(d_k, x) \quad \text{and} \quad w_\lambda(x) = \left(\max_{m=k^*+1,...,K} u(d_m, x) - \lambda \ell_{d_m} \right) - (\bar{u}(x) - \lambda \ell_{d_{k^*}}).$$

For each $x \in X$ there exists $m^*(x) \in \{k^*+1, ..., K\}$ such that

$$\begin{aligned} w_\lambda(x) &= u(d_{m^*(x)}, x) - \lambda \ell_{d_{m^*(x)}} + \lambda \ell_{d_{k^*}} - \bar{u}(x) \\ &< u(d_{m^*(x)}, x) - \lambda' \ell_{d_{m^*(x)}} + \lambda' \ell_{d_{k^*}} - \bar{u}(x) \leq w_{\lambda'}(x), \end{aligned}$$

where the strict inequality follows from $\lambda' < \lambda$ and $\ell_{d_{k^*}} < \ell_{d_{m^*(x)}}$. We thus obtain

$$w_\lambda(x) < w_{\lambda'}(x) \quad \text{for all } x \in X \text{ and all } \lambda' < \lambda. \tag{23}$$

Next, by (17) and (22), we have

$$x \in X^\lambda_{k^*} \iff w_\lambda(x) < 0. \tag{24}$$

Fix $\lambda' < \lambda$. To show (19), we construct an open set in which $w_{\lambda'}(x) \geq 0 > w_\lambda(x)$. If $X^{\lambda'}_{k^*}$ has measure zero, then the result follows. So we assume that $X^{\lambda'}_{k^*}$ has positive measure. By assumption, $X - X^\lambda_{k^*}$ also has a positive measure (note that by Assumption (A_2) the set $w_\lambda = 0$ has zero measure). This also implies that the set $\{x : w_\lambda(x) > 0\}$ has a

positive measure. Since X_k^λ is defined by a strict inequality between continuous functions by assumption (A_1), both sets have nonempty interiors. Let

$$y \in Int(X_{k^*}^{\lambda'}) \text{ and } z \in \{x : w_\lambda(x) > 0\} \cap Int(X).$$

Since X is convex and points y and z are in $Int(X)$, we have $\alpha y + (1-\alpha)z \in Int(X)$ for all $\alpha \in (0,1)$. Now, let

$$f(\alpha) = w_\lambda(\alpha y + (1-\alpha)z) \text{ and } g(\alpha) = w_{\lambda'}(\alpha y + (1-\alpha)z).$$

By (23), $f(\alpha) < g(\alpha)$ for all α, and $f(0) > 0$ and $g(1) < 0$. Let $\bar{\alpha} = \inf\{\alpha : g(\alpha) \leq 0\}$; then $g(\bar{\alpha}) = 0 > f(\bar{\alpha})$. Take $\alpha^* < \bar{\alpha}$ to be arbitrarily close to $\bar{\alpha}$ so that $f(\alpha^*) < 0 < g(\alpha^*)$, which exists by continuity, and let $x^* = \alpha^* y + (1-\alpha^*)z$, so $w_{\lambda'}(x^*) > 0 > w_\lambda(x^*)$. Since $w_{\lambda'}(x)$ and $w_\lambda(x)$ are continuous in x by (A_1), there exists an open neighborhood O_{x^*} of x^* such that $w_{\lambda'}(x) > 0 > w_\lambda(x)$ for all $x \in O_{x^*}$. By (24), this proves (19). □

Proof of Theorem 5.1. Let Q be an optimal inquiry under λ and let (D, ℓ) be its form. Let the actions in $D = \{d_1, ..., d_K\}$ be ordered according to (8). Then, by Corollary 4.2, the actions in D are ordered according to (16). Moreover, by Claim A.3, every category of Q has a positive probability, and so for $k = 1, ..., K-1$, both E_k^Q and $X - E_k^Q$ have positive measure. By Assumption (A_2) and absolute continuity of G, E_k^Q differs from X_k^λ only on a measure-zero set. Thus, the assumptions of Lemma A.1 with λ are satisfied. Next, by (5), (17), and the definition of E_k^Q and E_k^0 in Section 5, we obtain that, for each $k = 1, ..., K-1$, the event X_k^λ coincides with the set E_k^Q except, possibly, on a measure zero set. In words, conditional on event X_k^λ, inquiry Q almost surely leads to an action in $\{d_1, ..., d_k\}$. At $\lambda = 0$, the event X_k^0 contains the set E_k^0 up to a measure zero set. Then, Theorem 5.1 is immediate by Definition 5.1 and Lemma A.1 with $\lambda' = 0$. The case for $k = K$ is obvious as $E_K^Q = X$.

The fact that endogenous confirmation is strict when ℓ is not uniform also directly follows from Lemma A.1. When ℓ is uniform but $D \neq A^*$, an optimal uniform inquiry must omit at least one member of A^*, and hence $\mathbb{P}(E_K^Q) = 1 > \mathbb{P}(E_K^0)$. The result follows from $E_K^0 \subset E_K^Q$ from above and absolute continuity of G. □

Proof of Proposition 5.1. We first construct an open and dense set of costs such that, for each λ in the set, there is an $\epsilon_0 > 0$ and a form (D, ℓ) that is optimal for all $\lambda' \in [\lambda - \epsilon_0, \lambda + \epsilon_0]$.

Let $W^*(\lambda) = \max_{(D,\ell)\in\mathcal{F}^*} W(D,\ell,I^*(D,\ell))$. For each $(D,\ell)\in\mathcal{F}^*$, define

$$\Lambda(D,\ell) = \{\lambda > 0 : W(D,\ell,I^*(D,\ell);\lambda) = W^*(\lambda)\}.$$

Clearly, $\mathbb{R}_+ = \bigcup_{(D,\ell)\in\mathcal{F}^*} \Lambda(D,\ell)$. Note that each $\Lambda(D,\ell)$ is a closed set by Theorem 4.1 and the continuity of W. Moreover, because $\mathcal{F}^*$ is finite, the set $\Lambda_0 = \bigcup_{(D,\ell)\in\mathcal{F}^*} \text{int}(\Lambda(D,\ell))$ is open and dense over $\mathbb{R}_+$, and for each $\lambda\in\Lambda_0$, there exists a form $(D,\ell)\in\mathcal{F}^*$ and an interval $[\lambda-\epsilon_0,\lambda+\epsilon_0]$ with $\epsilon_0>0$, such that (D,ℓ) is an optimal form for any cost in that interval.

To show that the set $E_k^{Q^{\lambda'}}$ expands, we refine the above generic set so that the order (8) remains constant as well. For each given form and pair of actions, d, d', the choice probabilities (under the associated optimal categories) $\mathbb{P}(I_d^\lambda)$ are continuous in λ; hence the boundary of the set

$$\{\lambda > 0 : \mathbb{P}(I_d^\lambda) = \mathbb{P}(I_{d'}^\lambda)\}$$

is nowhere dense. Since there are finitely many forms and pairs, the set Λ_1 obtained from Λ_0 by excluding all these boundaries still leaves an open dense set. At every $\lambda\in\Lambda_1$, there is an $\epsilon>0$ such that not only the form of the optimal inquiry remains constant but also the set of orders satisfying (8) over the interval $[\lambda-\epsilon,\lambda+\epsilon]$.

Fix any form (D,ℓ) that is optimal throughout $[\lambda-\epsilon,\lambda+\epsilon]$, and let the actions in $D=\{d_1,...,d_K\}$ be ordered according to (8). Now consider arbitrary $\lambda_1<\lambda_2\in[\lambda-\epsilon,\lambda+\epsilon]$. Because (D,ℓ) is optimal under λ_2 and $\lambda_2>0$, by Corollary 4.2, the actions in D are ordered according to (16). As above, Claim A.3 implies the required positive-measure conditions. Thus, the assumptions of Lemma A.1 with $\lambda=\lambda_2$ are satisfied. Next, by (5), (17), and the definition of $E_k^{Q^\lambda}$ in Section 5, we obtain that, for each $k=1,...,K-1$, the event X_k^λ coincides with the set $E_k^{Q^\lambda}$ except, possibly, on a measure-zero set for $\lambda\in\{\lambda_1,\lambda_2\}$. Then, the result that endogenous confirmation is increasing is immediate by Definition 5.2 and Lemma A.1 where we set $\lambda=\lambda_2$ and $\lambda'=\lambda_1$.

Finally, to show strictness, consider the set Λ_2 obtained from Λ_1 by removing all boundary points of $\Lambda(D,\ell)$ for all $(D,\ell)\in\mathcal{F}^*$. Since these boundary points form a closed and nowhere dense set, Λ_2 is open and dense. Moreover, for all $\lambda\in\Lambda_2$, there is a neighborhood $[\lambda-\epsilon,\lambda+\epsilon]$ such that for each form (D,ℓ) that is optimal at λ, it is optimal over that neighborhood. Now, suppose that at λ there is an optimal form with a uniform length profile. Then, $W^*(\lambda')$ is linear over $[\lambda-\epsilon,\lambda+\epsilon]$. Since every form that is optimal at λ remains optimal over this

neighborhood, its value must coincide with W^*. By Lemma A.1, however, the value of a non-uniform form that remains optimal over an interval is strictly convex. Hence every optimal form at λ has a uniform length profile. Thus, if at λ one of the optimal forms is non-uniform, then all are non-uniform. Strictness follows directly from Lemma A.1. □

A.5. **Proof of Corollary 6.1.** Part (ii) follows immediately from Corollary 4.1. To prove part (i), let $\lambda > 0$, let Q^λ be an optimal inquiry under λ, let $(D^\lambda, \ell^\lambda, I^\lambda)$ be its outcome, and let $K = |A|$. If $D^\lambda \neq A$, then $\mathbb{P}(I^\lambda_d) > \mathbb{P}(I^\lambda_{d'}) = 0$ for $d \in D^\lambda$ and $d' \in A - D^\lambda$. If $D^\lambda = A$, then, by (4), ℓ^λ is not uniform. Thus, by Theorem 5.1, Q^λ has strict endogenous confirmation. That is, there exists an order $(d_k)_{k=1}^K$ that satisfies (8) and $k \in \{1, 2, ..., K\}$ such that $E_k^{Q^\lambda} \supseteq E_k^0$ and the set $E_k^{Q^\lambda} - E_k^0$ has a positive measure. It is then immediate that $\mathbb{P}(E_k^{Q^\lambda}) > \mathbb{P}(E_k^0)$, and thus $\mathbb{P}(I^\lambda_{d_1}) > \mathbb{P}(I^\lambda_{d_K})$. □

A.6. **Proof of Corollary 6.2.** Let Q be an optimal inquiry and let (D, ℓ, I) be its outcome. Denote $\delta(a', a'') = \sup_{x \in X} |u(a', x) - u(a'', x)|$. Suppose by contradiction that there exist $a', a'' \in D$ such that $\delta(a', a'') < \lambda$. There are two cases.

Case 1. Suppose that $\ell_{a'} \neq \ell_{a''}$. W.l.o.g., let $\ell_{a'} > \ell_{a''}$. Since (D, ℓ, I) is optimal, by Theorem 4.1, (5), and assumption (A_2), we have

$$I_{a'} = \{x \in X : u(a', x) > u(a, x) + \lambda(\ell_{a'} - \ell_a) \text{ for all } a \in D - \{a'\}\} \tag{25}$$

has a positive measure. Therefore, because $a'' \in D$, we must have

$$u(a', x) > u(a'', x) + \lambda(\ell_{a'} - \ell_{a''}) \geq u(a'', x) + \lambda \text{ for each } x \in I_{a'},$$

where the first inequality is by (25), and the second inequality is because $\ell_{a'} > \ell_{a''}$ and both $\ell_{a'}$ and $\ell_{a''}$ are integers. This contradicts the assumption that $\delta(a', a'') < \lambda$.

Case 2. Suppose that $\ell_{a'} = \ell_{a''}$. Consider an alternative outcome $(\hat{D}, \hat{\ell}, \hat{I})$ given by $\hat{D} = D - \{a''\}$, $\hat{\ell}_{a'} = \ell_{a'} - 1$, $\hat{\ell}_a = \ell_a$ for all $a \in D - \{a'\}$, $\hat{I}_{a'} = I_{a'} \cup I_{a''}$, and $\hat{I}_a = I_a$ for all $a \in D - \{a'\}$. In words, this outcome is the same as that of Q except that this outcome merges actions a' and a'' and the two categories that distinguish these actions into one. Because

$$\ell_{a'} = \ell_{a''} = \hat{\ell}_{a'} + 1, \tag{26}$$

we obtain $2^{-\ell_{a'}} + 2^{-\ell_{a''}} = 2^{-\hat{\ell}_{a'}}$. Since $\sum_{d\in D} 2^{-\ell_d} = 1$, we obtain that

$$\sum_{d\in\hat{D}} 2^{-\hat{\ell}_d} = \left(\sum_{d\in D-\{a',a''\}} 2^{-\hat{\ell}_d}\right) + 2^{-\hat{\ell}_{a'}} = \left(\sum_{d\in D-\{a',a''\}} 2^{-\ell_d}\right) + 2^{-\ell_{a'}} + 2^{-\ell_{a''}} = \sum_{d\in D} 2^{-\ell_d} = 1.$$

Thus, by Lemma 3.2, there exists an inquiry $\hat{Q}$ with outcome $(\hat{D}, \hat{\ell}, \hat{I})$. We obtain

$$\begin{aligned} W(\hat{Q};\lambda) - W(Q;\lambda) &= \int_{I_{a'}} \Big((u(a',x) - \lambda\hat{\ell}_{a'}) - (u(a',x) - \lambda\ell_{a'})\Big)G(\mathrm{d}x) \\ &\quad + \int_{I_{a''}} \Big((u(a',x) - \lambda\hat{\ell}_{a'}) - (u(a'',x) - \lambda\ell_{a''})\Big)G(\mathrm{d}x) \\ &= \int_{I_{a'}} \lambda G(\mathrm{d}x) + \int_{x\in I_{a''}} \Big(u(a',x) - u(a'',x) + \lambda\Big)G(\mathrm{d}x) > 0, \end{aligned}$$

where the first equality is by (2) and the fact that Q and $\hat{Q}$ differ only for $x \in I_{a'} \cup I_{a''}$, the second equality is by (26), and the inequality is because $\delta(a', a'') < \lambda$ and $I_{a'} \cup I_{a''}$ has a positive measure. We thus obtain a contradiction to the optimality of Q. □

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